

Aaron Xiong

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EDUCATION

University of Michigan

Bachelor of Science in Engineering in Computer Science, Minor in Electrical Engineering

Cumulative GPA: 3.92 /4; James B. Angell Scholar

Courses: Computer Architecture, Machine Learning, Embedded Systems, Computer Security, Data Structures and Algorithms.

Ann Arbor, MI, USA

Aug. 2024 – May. 2027

(Expected)

SKILLS

- **Languages & HDL:** C/C++, Python, JavaScript, MATLAB, SQL, SystemVerilog, ARMv7 Assembly, HTML/CSS, Swift
- **Libraries & Frameworks:** React, Flask, Tailwind CSS, Bootstrap, PyTorch, NumPy, ESP-IDF, Matplotlib, GTK, libcurl
- **Tools & Platforms:** Linux/Ubuntu, Git, Verdi, STM32, Arduino, Docker, Oracle Database, VMware, GDB, Wireshark, Ghidra

WORK EXPERIENCE

Full-stack Developer for U-M School of Nursing

Jan. 2025 – Dec. 2025

- Designed a HIPAA-compliant, mobile-friendly, full-stack web platform with a team of seven that aims to provide Nurse Health Coaching, Educational Content, and Personalized Community Resources for pregnant people and infant caregivers.
- Integrated Zoom and Qualtrics for nurse coaching and patient assessments, Google Maps for geospatial analysis, **Oracle Database** for patient data storage, and U-M's text-based **Generative AI** tool Maizey for summarization functionalities.
- Engineered and optimized prompts for Maizey to accurately summarize meeting notes from nurse-patient Zoom sessions, generate personalized care plans for clients, and extract key needs from health surveys while mitigating hallucinations to ensure safety.
- Designed an accessible and intuitive UI with **Figma** through interviews and usability testing, ensuring a seamless user experience.

PROJECT EXPERIENCE

Voxband – the ISC Lab (Embedded Systems)

Jan. 2026 – Present

- Engineered a wearable wristband integrating VPUs and air microphones to capture and filter **body-conducted speech signals**.
- Designed, manufactured, and assembled **custom multi-layer PCBs** and 3D-printed enclosures to deploy the experimental prototype.
- Conducted spectral analysis and noise-source isolation tests to debug sources of audio interference and improve signal quality.
- Programmed firmware to support **4-channel USB audio streaming**, Wi-Fi connections, and EMMC-based memory management.
- Built signal-processing and evaluation pipelines to benchmark **speaker diarization** robustness under diverse acoustic conditions.

Founder of RISC-M

Apr. 2026 – Present

- Founded the University of Michigan's first student organization dedicated to computer architecture, establishing a community of **40+ students** to engage in open-source projects, research discussions, career outreach, and architecture design competitions.
- Defined the organization's operational roadmap and managed administrative coordination, communications, and logistics.
- Initiated and led a paper reading group, collaborating with faculty to present and discuss novel computer architecture research.
- Coordinated guest lectures and networking events, inviting industry engineers to give technical talks and share career insights.

Superscalar Processor Design (SystemVerilog)

Mar. 2026 – Apr. 2026

- Designed a 2-way superscalar, 4-stage, out-of-order RISC processor for the LC2K ISA and verified pipeline execution in **Verdi**.
- Implemented a parameterizable **micro-op queue** with a tree priority selector for hazard-aware out-of-order dispatch scheduling.
- Designed a set-associative instruction cache with an area-efficient banked branch predictor and branch target buffer, improving instruction-fetch efficiency and branch prediction accuracy to **reduce average CPI from 2.06 to 1.60**.
- Evaluated architectural tradeoffs by benchmarking module sizes and constructing Pareto analyses for PPA optimization.
- Managed team integration and debugging workflows through milestone tracking, diagram-first design, and atomic commits.

Site-Invariant Brain Age Estimator (ML, Python)

Jan. 2026 – Apr. 2026

- Designed a site-invariant ML pipeline to predict chronological brain age from structural MRI scans across 64 scanner configurations.
- Implemented a **3D ResNet-18** backbone with Gradient Reversal Layer-based domain-adversarial training to mitigate scanner-induced domain shift by learning site-invariant features, reducing held-out scanner generalization error **from 2.99 to 1.24 years**.
- Stabilized adversarial training by integrating a spectrally-normalized discriminator, label smoothing, and feature noise injection.
- Improved MAE **from 4.9 to 3.8 years** by integrating heteroscedastic **deep ensembles** for out-of-distribution sample filtering.

PUBLICATIONS

- Zou, K., **Xiong, A.**, Zhang, Y., Zhang, F., Ren, Y., Yang, J., Lee, A., Bhushan, S., & Wang, L. "LiveOIBench: Can Large Language Models Outperform Human Contestants in Informatics Olympiads?" Accepted to **ICML 2026**.